Machine Learning for Recognizing Epileptic Seizures in EEG Recordings

A comparative study of five classifiers on the Bonn EEG dataset

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ABSTRACT

Background. Epilepsy affects around 50 million people worldwide, and electroencephalography (EEG) is central to its diagnosis. Reviewing EEG for seizure activity is largely manual and time-consuming, which motivates automated methods based on machine learning.

Objective. To build and evaluate machine-learning models that recognize seizure activity and epilepsy-related patterns in EEG, and to identify which signal characteristics and evaluation choices drive their performance.

Methods. Five hundred single-channel recordings from the public Bonn EEG dataset were band-pass filtered (0.5–40 Hz), segmented into 5,500 two-second windows and described by 18 time- and frequency-domain features. Five classifiers (logistic regression, support vector machine, random forest, gradient boosting and a multilayer perceptron) were trained on six classification tasks and evaluated with recording-grouped 5-fold cross-validation repeated three times, so that no test recording was seen during training.

Results. Seizure versus all non-seizure EEG was recognized with 96.6–97.5% balanced accuracy per window (97.0–98.3% per recording; AUC 0.995–0.998). Seizure versus healthy EEG reached 99.5–100%, but signal amplitude alone achieved 99.7% on the same task. Distinguishing the epileptogenic zone from the opposite hemisphere between seizures was harder (65.6–71.8% per window; up to 81.3% per recording) and continued to improve as training data increased. Window-level shuffling inflated accuracy by up to 3.1 points, and limiting the variety of non-seizure training data raised false alarms on patient EEG to as much as 32.2%.

Conclusions. Standard classifiers recognize seizure activity in this dataset with high accuracy, largely on the basis of signal amplitude. Evaluation protocol, training-data composition and recording setup materially affect measured performance and should be reported alongside accuracy.

Keywords: EEG; epilepsy; seizure recognition; machine learning; feature engineering; cross-validation; data leakage

Disclaimer

This is an exploratory study on a public research dataset. The models have not been clinically validated and must not be used for diagnosis or medical decision-making.

Highlights

- Five classifiers recognized seizure EEG with 96.6–97.5% balanced accuracy on recordings unseen during training.
- On seizure versus healthy EEG, amplitude alone matched the full models (99.7%), showing that a high score does not imply rich learned structure.
- Separating the epileptogenic zone from the opposite hemisphere between seizures was markedly harder (up to 81.3% per recording) and data-limited.
- Splitting data by window instead of by recording inflated accuracy by up to 3.1 points; narrow non-seizure training data raised false alarms to 32.2%.
- Because healthy and patient EEG were recorded differently (scalp vs intracranial), healthy-versus-epilepsy results cannot be attributed to epilepsy alone.

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1. Introduction

1.1 Problem statement

Epilepsy is a neurological condition characterized by a tendency to have recurrent seizures, and it affects around 50 million people worldwide (World Health Organization). Electroencephalography (EEG), which records the brain's electrical activity, is one of the main tools used to identify seizure activity and to locate where seizures begin. Interpreting EEG, however, is largely a manual task: trained specialists visually review recordings that can last hours or days.

Machine learning offers a way to automate part of this review. Instead of writing explicit rules, a model learns to map EEG measurements to labels such as “seizure” or “no seizure” from examples whose labels are already known. The public Bonn EEG dataset (Andrzejak et al., 2001) has become a common starting point for developing such methods, because it provides clean, labeled segments from healthy volunteers and from patients with epilepsy, both during and between seizures.

A model's usefulness depends not only on its accuracy, but on what that accuracy reflects. This study therefore builds a complete machine-learning pipeline for seizure recognition and examines, in detail, which signal characteristics and evaluation choices drive the results.

1.2 Objectives

1. Build an end-to-end pipeline that turns raw EEG into features and trains classifiers to recognize seizure activity.
2. Compare five widely used classifiers across six tasks of increasing difficulty, evaluated only on recordings unseen during training.
3. Identify the signal characteristics the models rely on.
4. Quantify how the evaluation protocol, the composition of the training data and the amount of training data affect measured performance.
5. Document the method so it can be reproduced and applied to other single-channel EEG datasets.

1.3 Scope

The models classify short, pre-labeled EEG segments. They do not forecast seizures before onset, and they do not distinguish a single (provoked) seizure from epilepsy. Evaluation is performed on recordings unseen during training; because the dataset does not identify which recording belongs to which person, performance on entirely new individuals is not assessed.

1.4 Organization

Section 2 introduces the relevant background on EEG and machine learning. Section 3 describes the data, preprocessing, features, models and evaluation protocol. Section 4 reports the results, Section 5 discusses their implications and limitations, and Section 6 concludes. Appendix A provides a step-by-step guide to applying the pipeline to new data.

2. Background: EEG and machine learning

2.1 The brain's electrical activity

The human brain contains approximately 86 billion neurons (Azevedo et al., 2009), which communicate through electrical impulses and chemical signals. A single neuron's activity is far too small to detect from

outside the head, but when large populations of neurons are active in synchrony, their summed electrical fields become measurable.

EEG measures these fields with electrodes. In standard (scalp) EEG, electrodes are placed on the scalp and record voltage fluctuations on the order of microvolts. Hans Berger recorded the first human EEG in 1924 (Berger, 1929). In some clinical settings, notably the evaluation of patients for epilepsy surgery, electrodes are placed inside the skull, directly on or within brain tissue. This **intracranial EEG** records more localized activity because the signal does not pass through the skull and scalp. To a computer, an EEG channel is simply a time series: a sequence of voltage values sampled at a fixed rate.

2.2 Brain rhythms

EEG is commonly described in terms of oscillations at different frequencies, grouped into conventional bands (Table 1, Figure 1). The balance between these rhythms changes with the state of the brain, from deep sleep to focused attention.

Table 1. Conventional EEG frequency bands.

Band	Frequency	Typically associated with
Delta	0.5–4 Hz	Deep sleep
Theta	4–8 Hz	Drowsiness, light sleep
Alpha	8–13 Hz	Relaxed wakefulness, especially with eyes closed
Beta	13–30 Hz	Alertness, active thinking
Gamma	> 30 Hz	Intensive information processing

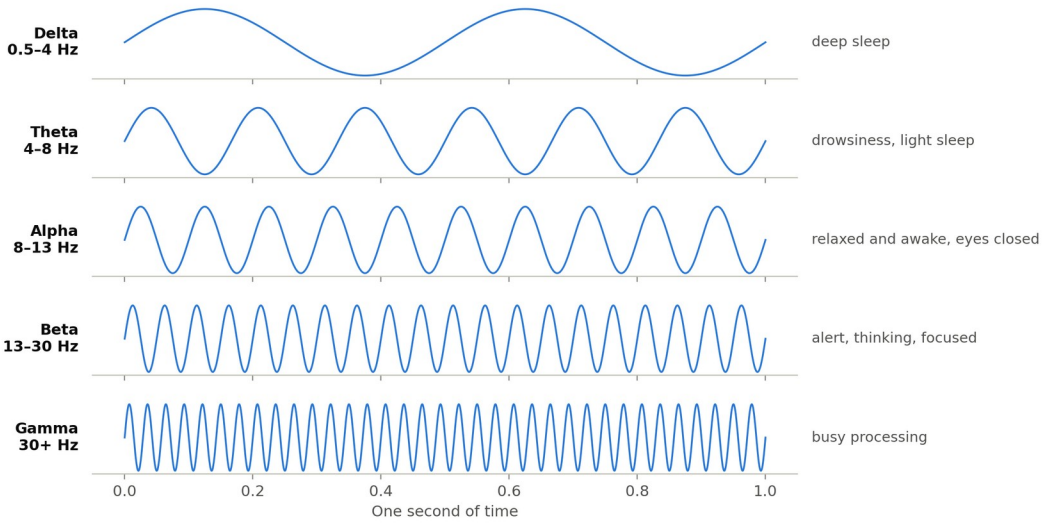


Figure 1. Illustration of the five conventional EEG bands as pure sine waves over one second. Real EEG is a mixture of many such components.

2.3 Seizures and epilepsy

A **seizure** is a transient episode of abnormal, excessive and synchronized neuronal activity. On EEG, seizure activity typically appears as high-amplitude, rhythmic or spiky waveforms. Clinical EEG terminology distinguishes the **ictal** state (during a seizure) from the **interictal** state (between seizures).

The **epileptogenic zone** is the region of the brain from which seizures originate; the **contralateral** side is the corresponding region in the opposite hemisphere.

2.4 Machine learning fundamentals

Machine learning is a branch of artificial intelligence in which a computer learns a decision rule from data rather than following rules written by a person. This study uses **supervised learning**: each training example comes with its correct answer, or **label**. Each example is described by **features**, numerical measurements that summarize it. In this work an example is a two-second EEG window, its label is the group it came from, and its features include measures such as amplitude and rhythmic regularity.

Training is the process of adjusting a model's internal parameters so that its predictions on the training examples match their labels as closely as possible. **Inference** is the use of the trained, now-fixed model to predict labels for new examples. The two phases must apply identical preprocessing (Figure 2).

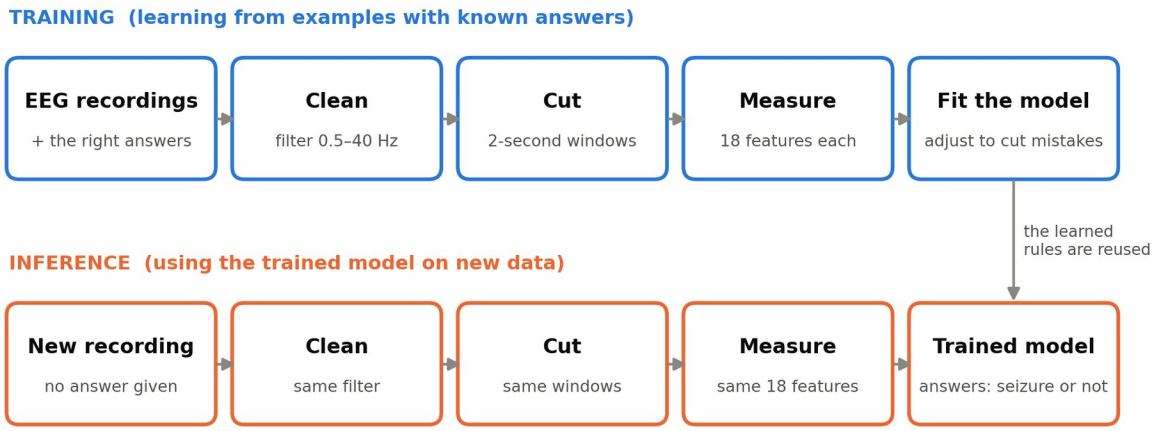


Figure 2. Training and inference. During training (top), labeled recordings are cleaned, segmented and converted into features, and the model's parameters are fitted. During inference (bottom), new recordings pass through the identical steps, and the trained model outputs a prediction.

Table 2 illustrates the idea of training with a deliberately simple model: a single threshold on one feature. Starting from a poor threshold, the learning procedure adjusts it to reduce the number of misclassified examples. Practical algorithms perform the same kind of optimization automatically, over many features and thousands of examples.

Table 2. Illustrative example: learning a threshold on wave amplitude (synthetic values).

Example	Amplitude (μV)	Label	Rule “> 500”	Rule “> 200”	Rule “> 100”
1	40	No seizure	✓	✓	✓
2	55	No seizure	✓	✓	✓
3	60	No seizure	✓	✓	✓
4	180	Seizure	✗	✗	✓
5	250	Seizure	✗	✓	✓
6	300	Seizure	✗	✓	✓
		Errors	**3**	**1**	**0**

A model that performs well on its training data may still fail on new data if it has memorized idiosyncrasies of the training set, a problem known as **overfitting**. For this reason, performance must always be measured on data held out from training. Table 3 summarizes the evaluation metrics used in this study.

Table 3. Evaluation metrics.

Metric	Definition	Interpretation
Accuracy	Share of all predictions that are correct	Misleading when classes are imbalanced
Balanced accuracy	Mean of the per-class accuracies	Main metric here; 50% is chance for two classes
Sensitivity	Share of true seizure windows that are detected	How many seizures are caught
Specificity	Share of non-seizure windows correctly rejected	1 − specificity is the false-alarm rate
Confusion matrix	Counts of predicted vs true class	Shows which classes are confused
ROC curve, AUC	Sensitivity vs false-alarm rate across decision thresholds; area under it	1.0 is perfect ranking; 0.5 is chance

2.5 Classifiers

Five classifiers representing the main families of classical machine learning were compared (Table 4). Using several models makes it possible to separate effects of the data from effects of a particular algorithm.

Table 4. Classifiers compared in this study.

Classifier	Principle	Intuition
Logistic regression	Weighted linear combination of features passed through a logistic function	A scoring checklist with a cut-off
Support vector machine (SVM)	Maximum-margin boundary; an RBF kernel allows non-linear boundaries	The widest possible gap between classes
Random forest	Ensemble of decision trees trained on bootstrap samples; majority vote	Many independent experts voting
Gradient boosting	Trees added sequentially, each correcting the errors of the ensemble so far	Each new member fixes remaining mistakes
Multilayer perceptron (MLP)	Feed-forward neural network with two hidden layers	Layers of simple units learning feature combinations

3. Materials and methods

3.1 Dataset

The study uses the Bonn EEG dataset (Andrzejak et al., 2001), which is freely available for research and education. It contains five sets of 100 single-channel EEG segments, each 23.6 s long (Tables 5 and 6). Sets A

and B were recorded from five healthy volunteers using scalp electrodes; sets C, D and E were recorded intracranially from five patients with epilepsy.

Table 5. Composition of the Bonn EEG dataset.

Set	Subjects and recording site	Electrodes	State	Segments
A	Healthy volunteers, eyes open	Scalp	Normal	100
B	Healthy volunteers, eyes closed	Scalp	Normal	100
C	Patients; hippocampal formation of the hemisphere opposite the epileptogenic zone	Intracranial	Interictal	100
D	Patients; within the epileptogenic zone	Intracranial	Interictal	100
E	Patients; seizure activity	Intracranial	Ictal	100

Table 6. Technical characteristics.

Property	Value
Segments	500 (5 sets × 100)
Duration	23.6 s (4,097 samples)
Sampling rate	173.61 Hz
Channels per segment	1
Acquisition	128-channel system, 12-bit analog-to-digital conversion
Selection	Segments selected from longer recordings and screened for artifacts such as eye movements

3.2 Preprocessing

Each segment was filtered with a fourth-order zero-phase Butterworth band-pass filter (0.5–40 Hz) to remove slow drift and high-frequency noise. Filtered segments were divided into non-overlapping two-second windows (347 samples), yielding 11 windows per segment and 5,500 windows in total. Each window retained the identifier of its source recording, which is required for grouped evaluation (Section 3.6).

3.3 Feature extraction

Each window was summarized by 18 features (Table 7). Four **amplitude** features depend on the overall signal size; fourteen **shape** features are invariant to scaling and describe spectral content, regularity and waveform morphology. Spectral features were computed from Welch power spectral density estimates restricted to 0.5–40 Hz. Hjorth parameters follow Hjorth (1970) and permutation entropy (order 3) follows Bandt and Pompe (2002).

Table 7. Features extracted from each two-second window.

Feature	Family	Description
Standard deviation (log)	Amplitude	Overall size of the signal
Line length (log)	Amplitude	Mean absolute sample-to-sample change
Peak-to-peak range (log)	Amplitude	Difference between maximum and minimum
Total power (log)	Amplitude	Spectral power in 0.5–40 Hz
Relative band power (×5)	Shape	Share of power in delta, theta, alpha, beta and gamma bands
Peak frequency	Shape	Frequency of maximum power
Spectral edge frequency (90%)	Shape	Frequency below which 90% of power lies
Spectral entropy	Shape	Flatness of the power spectrum
Hjorth mobility	Shape	Proxy for mean frequency
Hjorth complexity	Shape	Deviation from a pure sine wave
Zero-crossing rate	Shape	Rate of sign changes
Skewness	Shape	Asymmetry of the amplitude distribution
Kurtosis	Shape	Peakedness of the amplitude distribution
Permutation entropy	Shape	Irregularity of ordinal patterns (0 = regular, 1 = random)

3.4 Classification tasks

Six tasks of increasing difficulty were defined (Table 8). T3 and T5 compare only intracranial recordings and are therefore not affected by the difference in electrode placement between healthy and patient data.

Table 8. Classification tasks.

Task	Classes	Purpose
T1 Seizure vs healthy	A vs E	Common introductory task
T2 Seizure vs all non-seizure	A, B, C, D vs E	Realistic seizure recognition
T3 Seizure vs interictal	C, D vs E	Seizure recognition within patients (intracranial only)
T4 Healthy vs interictal	A, B vs C, D	Epilepsy-related patterns without seizures
T5 Epileptogenic zone vs contralateral	C vs D	Localization-related differences (intracranial only)
T6 Five-class	A, B, C, D, E	Full discrimination (chance = 20%)

3.5 Models and training

All models were implemented in scikit-learn (Pedregosa et al., 2011) with fixed settings and no hyperparameter tuning, so that differences reflect the algorithms rather than tuning effort. Features were standardized for logistic regression, the SVM and the MLP; class weights were balanced for all models except the MLP. Table 9 lists the configurations and the size of each model when trained on all 5,500 windows for task T2.

Table 9. Model configurations and trained-model complexity (task T2, all data).

Model	Configuration	Learned structure	Training time
Logistic regression	Features rescaled; groups balanced	19 numbers (one weight per feature + 1)	< 0.1 s
SVM (RBF kernel)	Curved (RBF) boundary; C = 10; groups balanced	Kept 215 training windows as 'support vectors'	< 0.1 s
Random forest	300 trees; groups balanced	300 trees, 37,904 yes/no nodes in total	1.7 s
Gradient boosting	Up to 100 trees added one at a time; learning rate 0.1	100 trees, 6,084 yes/no nodes in total	0.4 s
Neural network (MLP)	2 hidden layers (64, 32); stops early when it stops improving	3,329 connection strengths (layers 18 → 64 → 32 → 1)	0.6 s

Timings on a 2-core cloud machine. Inference required less than 0.03 ms per window for every model.

3.6 Evaluation protocol

Performance was estimated with **recording-grouped stratified 5-fold cross-validation**, repeated three times with different random splits (15 train–test splits per model and task). All windows from a given recording were assigned to the same fold, so every test recording was unseen during training. For task T2, each split trained on 400 recordings (4,400 windows) and tested on 100 (1,100 windows). Window-level predictions were also aggregated into a recording-level decision by majority vote over the recording's 11 windows. Results are reported as mean ± standard deviation across the 15 splits.

3.7 Additional analyses

- **Feature ablation:** gradient boosting trained on amplitude features only, shape features only, and the standard deviation alone.
- **Evaluation protocol:** all tasks repeated with window-level shuffling (no grouping), allowing windows of one recording to appear in both training and test sets.
- **Training-set composition:** three seizure detectors trained with different non-seizure classes (A only; A+B; A–D), each evaluated on the same held-out recordings from all five sets.
- **Feature importance:** permutation importance on held-out folds (10 permutations per fold) for tasks T3 and T5.
- **Learning curve:** gradient boosting trained on 10–100% of the available training recordings (class proportions preserved) for tasks T2 and T5.
- **Held-out inference:** a single model trained on 80% of recordings and applied to the remaining 20%, reported per recording.

3.8 Implementation

The pipeline was implemented in Python using NumPy, SciPy, scikit-learn, pandas and matplotlib, and was run both locally and in Google Colab. The main experiment trained approximately 1,500 models in about nine minutes on a standard computer; the additional analyses take about one further minute. Code and results are publicly available (see Availability).

4. Results

4.1 Signal characteristics

Figure 3 shows representative segments from each set. Ictal EEG (set E) is markedly larger and more irregular than all other sets. Figure 4 shows the average power spectra: set B shows a pronounced alpha peak near 10–11 Hz, consistent with eyes-closed rest, while the intracranial interictal sets (C, D) contain substantially less high-frequency power than the scalp recordings (A, B). Window amplitudes (Figure 5) had medians of 38, 55, 46 and 47 µV for sets A–D and 276 µV for set E.



Figure 3. Representative EEG from each set (first 8 s of one segment per set, identical vertical scale).

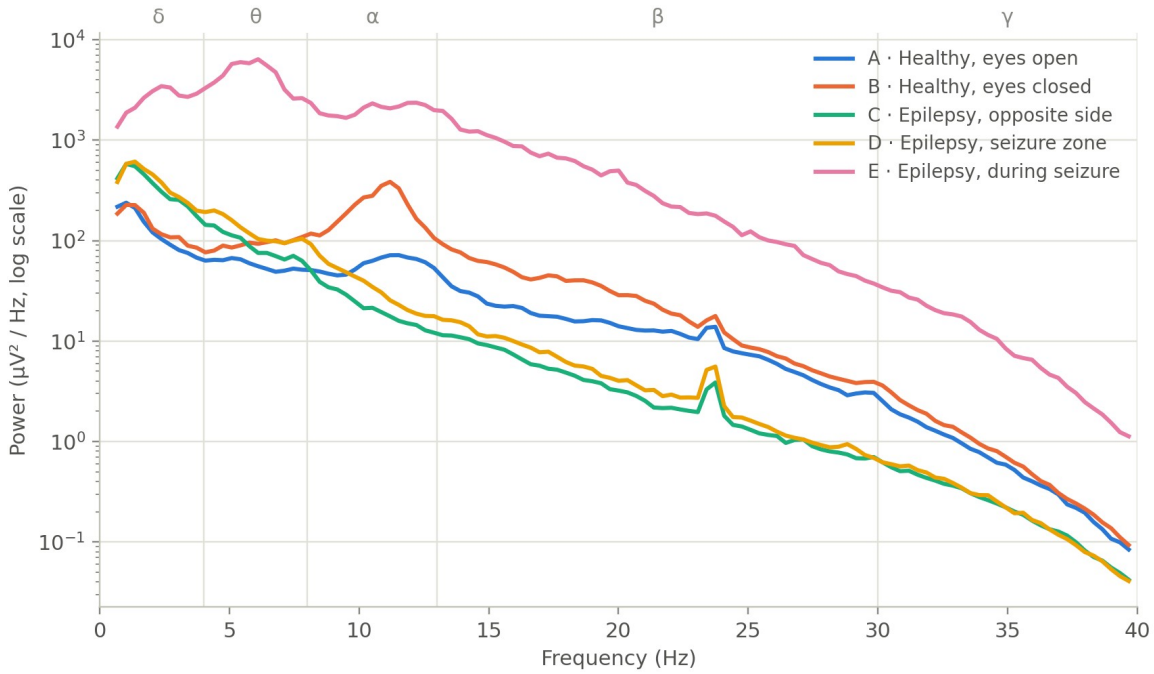


Figure 4. Mean power spectral density of each set (log scale), with conventional frequency bands marked.

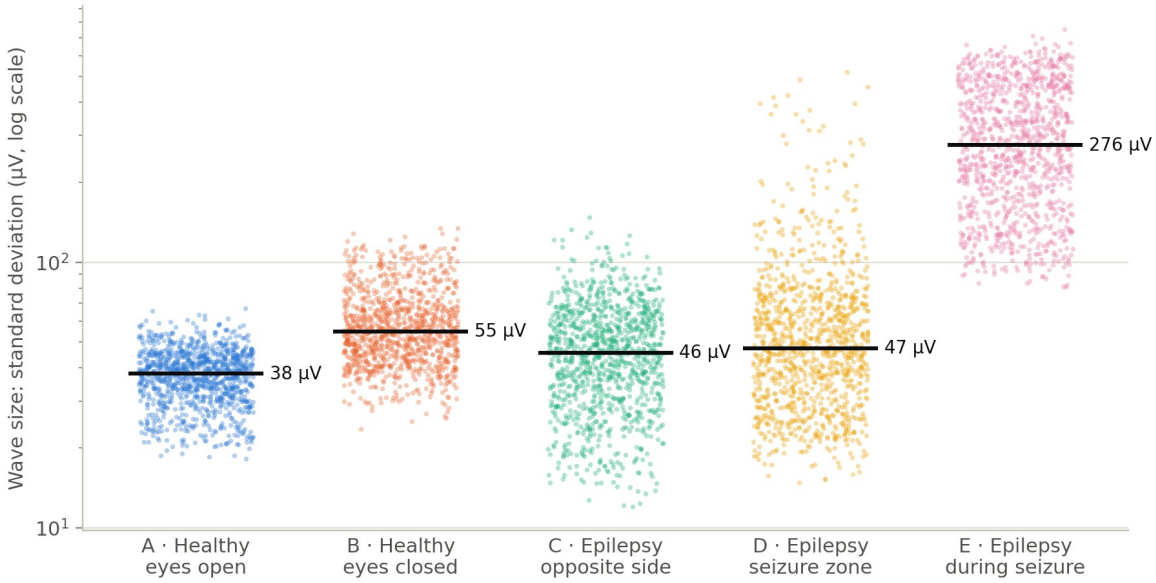


Figure 5. Distribution of window amplitude (standard deviation) by set; black lines mark medians. Ictal windows are roughly five to seven times larger than windows from any other set.

4.2 Classification performance

Table 10 and Figure 6 report window-level balanced accuracy for every model and task; Table 11 reports recording-level results. All five models achieved 94.5–100% on tasks T1–T4. On the harder tasks T5 and T6, the SVM and the two tree ensembles outperformed logistic regression and the MLP by 3–6 percentage

points. Recording-level aggregation improved accuracy on every task, and Table 12 summarizes the best classifier for each task.

Table 10. Window-level balanced accuracy (%), mean ± SD over 15 splits. Best model per task in bold.

Task	Logistic regression	SVM	Random forest	Gradient boosting	Neural network
T1 Seizure vs healthy	99.7 ± 0.5	99.7 ± 0.3	**100.0 ± 0.0**	99.9 ± 0.1	99.5 ± 0.7
T2 Seizure vs all non-seizure	**97.5 ± 1.4**	96.9 ± 1.2	96.7 ± 1.9	97.4 ± 1.6	96.6 ± 1.6
T3 Seizure vs interictal	96.8 ± 1.7	96.9 ± 1.7	96.9 ± 2.1	**97.3 ± 1.8**	96.4 ± 2.3
T4 Healthy vs interictal	94.5 ± 1.2	96.5 ± 0.8	96.0 ± 1.1	**96.7 ± 1.0**	96.0 ± 1.0
T5 Epileptogenic zone vs contralateral	65.9 ± 4.3	**71.8 ± 3.2**	71.5 ± 4.1	71.4 ± 2.8	65.6 ± 4.1
T6 Five-class	74.8 ± 1.9	78.1 ± 2.2	79.3 ± 2.5	**79.3 ± 1.8**	75.0 ± 2.4

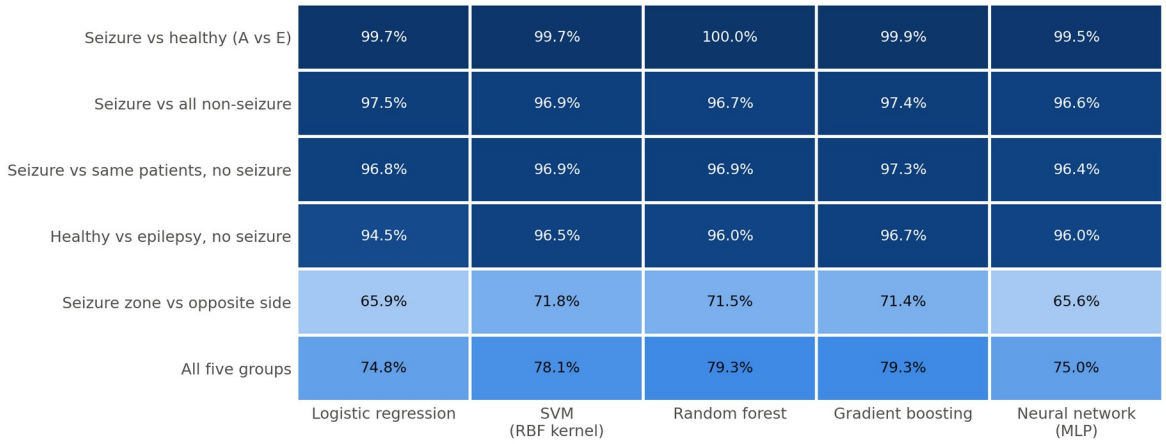


Figure 6. Window-level balanced accuracy of each classifier on each task (rows follow tasks T1–T6).

Table 11. Recording-level balanced accuracy (%), majority vote over 11 windows.

Task	Logistic regression	SVM	Random forest	Gradient boosting	Neural network
T1 Seizure vs healthy	99.8%	100.0%	100.0%	100.0%	100.0%
T2 Seizure vs all non-seizure	98.3%	97.4%	97.0%	98.1%	97.0%
T3 Seizure vs interictal	97.8%	97.8%	97.6%	97.8%	97.9%
T4 Healthy vs interictal	96.9%	99.0%	98.4%	99.2%	98.4%
T5 Epileptogenic zone vs contralateral	72.3%	81.3%	78.2%	79.7%	71.8%
T6 Five-class	81.4%	83.9%	85.8%	87.4%	80.6%

Table 12. Summary: best classifier per task.

Task	Best classifier	Window BA (%)	Recording BA (%)	AUC
T1 Seizure vs healthy	Random forest	100.0	100.0	1.000
T2 Seizure vs all non-seizure	Logistic regression	97.5	98.3	0.995
T3 Seizure vs interictal	Gradient boosting	97.3	97.8	0.997
T4 Healthy vs interictal	Gradient boosting	96.7	99.2	0.995
T5 Epileptogenic zone vs contralateral	SVM	71.8	81.3	0.794
T6 Five-class	Gradient boosting	79.3	87.4	—

4.3 Seizure recognition

Table 13 details the two seizure-recognition tasks. On T2, sensitivity ranged from 94.3% to 97.1% and specificity from 97.8% to 99.0%, with AUC between 0.995 and 0.998 (Figure 7 shows the ROC curve for gradient boosting). Performance on T3, where all recordings are intracranial, was similar (96.4–97.3% balanced accuracy), indicating that seizure recognition does not depend on the scalp–intracranial difference.

Table 13. Sensitivity, specificity and AUC for seizure recognition (window level).

Task	Classifier	Sensitivity	Specificity	AUC	Balanced acc.
T2	Logistic regression	97.1%	97.8%	0.995	97.5%
T2	SVM (RBF kernel)	95.0%	98.8%	0.997	96.9%
T2	Random forest	94.3%	99.0%	0.998	96.7%
T2	Gradient boosting	96.0%	98.9%	0.998	97.4%
T2	Neural network (MLP)	94.5%	98.7%	0.995	96.6%
T3	Logistic regression	96.8%	96.8%	0.995	96.8%
T3	SVM (RBF kernel)	95.3%	98.5%	0.997	96.9%
T3	Random forest	95.8%	98.0%	0.997	96.9%
T3	Gradient boosting	96.9%	97.8%	0.997	97.3%
T3	Neural network (MLP)	94.8%	97.9%	0.993	96.4%

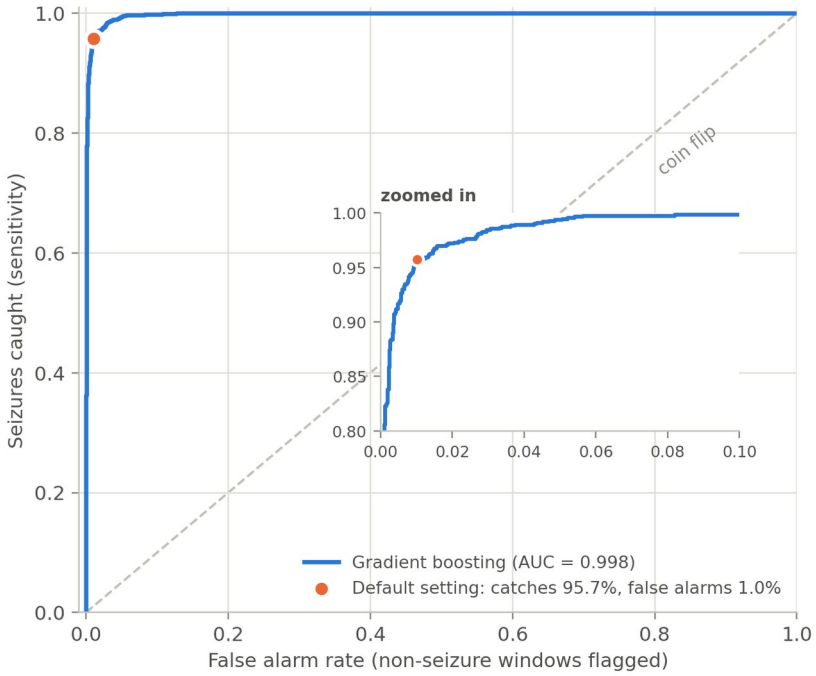


Figure 7. ROC curve for gradient boosting on task T2, from out-of-fold predictions (AUC = 0.998). The marker shows the default decision threshold of 0.5: sensitivity 95.7%, false-alarm rate 1.0%. Inset: magnified upper-left region.

4.4 Inference on held-out recordings

A gradient boosting model trained once on 400 recordings was applied to the remaining 100 recordings (1,100 windows). It achieved 99.3% window-level balanced accuracy and classified all 100 recordings correctly by majority vote (Table 14). This single split is more favorable than the cross-validated estimate for the same model (97.4% per window, 98.1% per recording), illustrating the value of repeated evaluation.

Table 14. Recording-level predictions on 100 held-out recordings (task T2, gradient boosting).

Set	Held-out recordings	Correctly classified
A (healthy, eyes open)	18	18
B (healthy, eyes closed)	21	21
C (interictal, contralateral)	19	19
D (interictal, epileptogenic zone)	17	17
E (ictal)	25	25
Total	**100**	**100**

4.5 Five-class discrimination

On the five-class task (chance level 20%), the best models reached 79.3% balanced accuracy per window and 87.4% per recording. Ictal windows were recognized in 96% of cases (Figure 8). Most errors occurred between physiologically similar pairs: eyes open versus eyes closed (A, B) and the two interictal sets (C, D), which were confused with each other in 27–28% of windows.

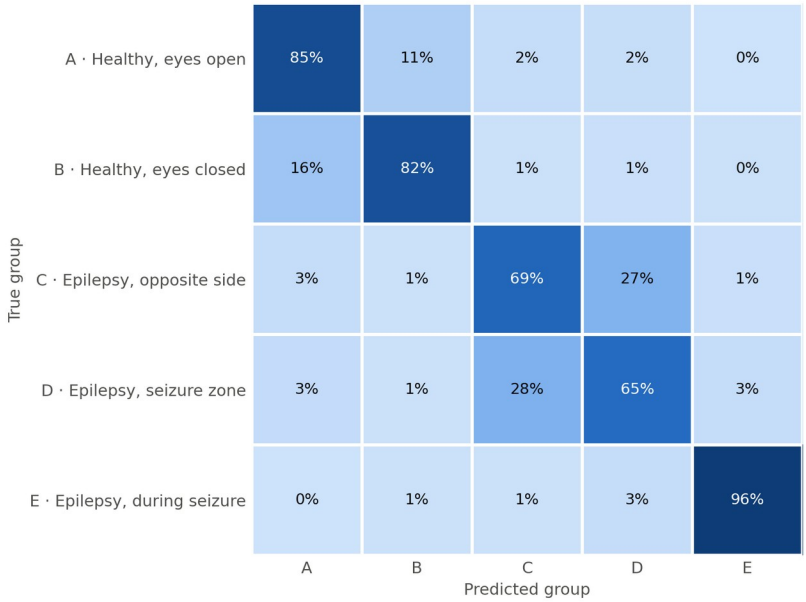


Figure 8. Five-class confusion matrix for gradient boosting (row-normalized, pooled over 15 splits).

4.6 Feature importance

Permutation importance (Figure 9) showed that seizure recognition (T3) relied predominantly on amplitude features, namely standard deviation and line length. Discrimination of the epileptogenic zone (T5) instead relied on line length, permutation entropy and Hjorth mobility. On average, windows from the epileptogenic zone had slightly greater line length and slightly lower permutation entropy, i.e. more regular ordinal patterns, than contralateral windows.

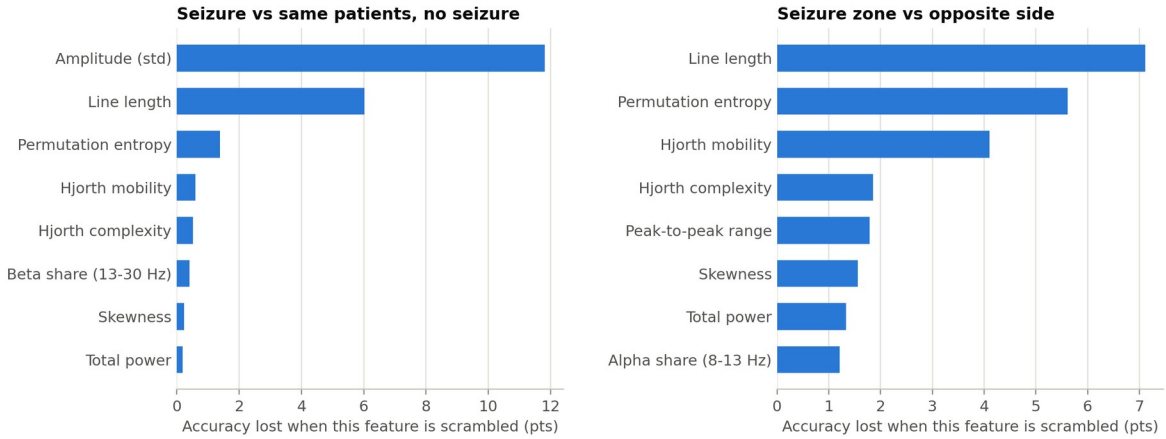


Figure 9. Permutation importance (decrease in balanced accuracy when a feature is shuffled) for gradient boosting on tasks T3 (left) and T5 (right); top eight features.

4.7 Feature ablation

Table 15 compares gradient boosting trained on subsets of features. The standard deviation alone achieved 99.7% on T1, equal to the full feature set within one standard deviation, and 96.0% on T2. On the harder tasks it fell to near chance (52.1% on T5), and the shape features provided most of the remaining discriminative information.

Table 15. Feature ablation: gradient boosting balanced accuracy (%) by feature subset.

Task	All (18)	Shape (14)	Amplitude (4)	Std. dev. only (1)
T1 Seizure vs healthy	99.9%	97.6%	99.9%	99.7%
T2 Seizure vs all non-seizure	97.4%	93.9%	96.1%	96.0%
T3 Seizure vs interictal	97.3%	95.0%	95.8%	95.0%
T4 Healthy vs interictal	96.7%	94.8%	93.8%	57.8%
T5 Epileptogenic zone vs contralateral	71.4%	65.3%	61.4%	52.1%
T6 Five-class	79.3%	73.3%	68.3%	47.6%

4.8 Effect of the evaluation protocol

Splitting by window rather than by recording increased the averaged balanced accuracy on every task (Figure 10, Table 16). The increase was small on easy tasks (+0.1 points on T1 and T4) and largest on the hardest tasks (+3.1 points on T5, +2.5 on T6).

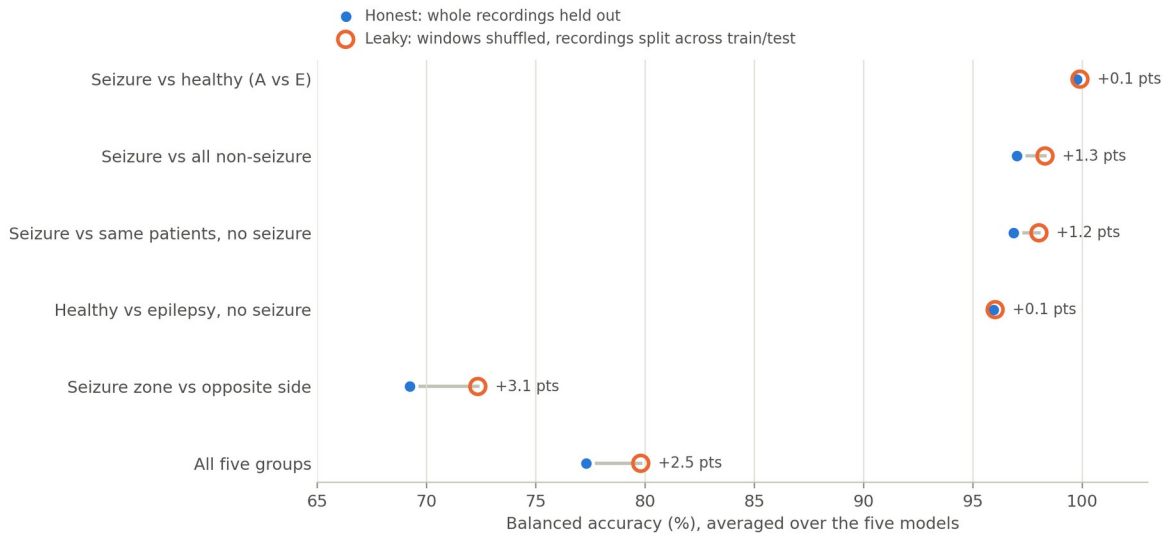


Figure 10. Balanced accuracy under recording-grouped (honest) and window-shuffled (leaky) evaluation, averaged over the five classifiers.

Table 16. Recording-grouped vs window-shuffled evaluation (mean of five classifiers).

Task	Grouped (%)	Shuffled (%)	Difference (points)
T1 Seizure vs healthy	99.8	99.9	+0.1
T2 Seizure vs all non-seizure	97.0	98.3	+1.3
T3 Seizure vs interictal	96.9	98.0	+1.2
T4 Healthy vs interictal	95.9	96.0	+0.1
T5 Epileptogenic zone vs contralateral	69.2	72.3	+3.1
T6 Five-class	77.3	79.8	+2.5

4.9 Effect of training-set composition

Detectors trained only on set A as the non-seizure class detected 99.7% of ictal windows but flagged 17.5% of eyes-closed windows (B), 18.7% of contralateral interictal windows (C) and 32.2% of epileptogenic-zone interictal windows (D) as seizures (Figure 11, Table 17). Including all non-seizure sets during training reduced these false-alarm rates to 0.2%, 1.1% and 3.8%, respectively, at the cost of a lower detection rate (95.5%).

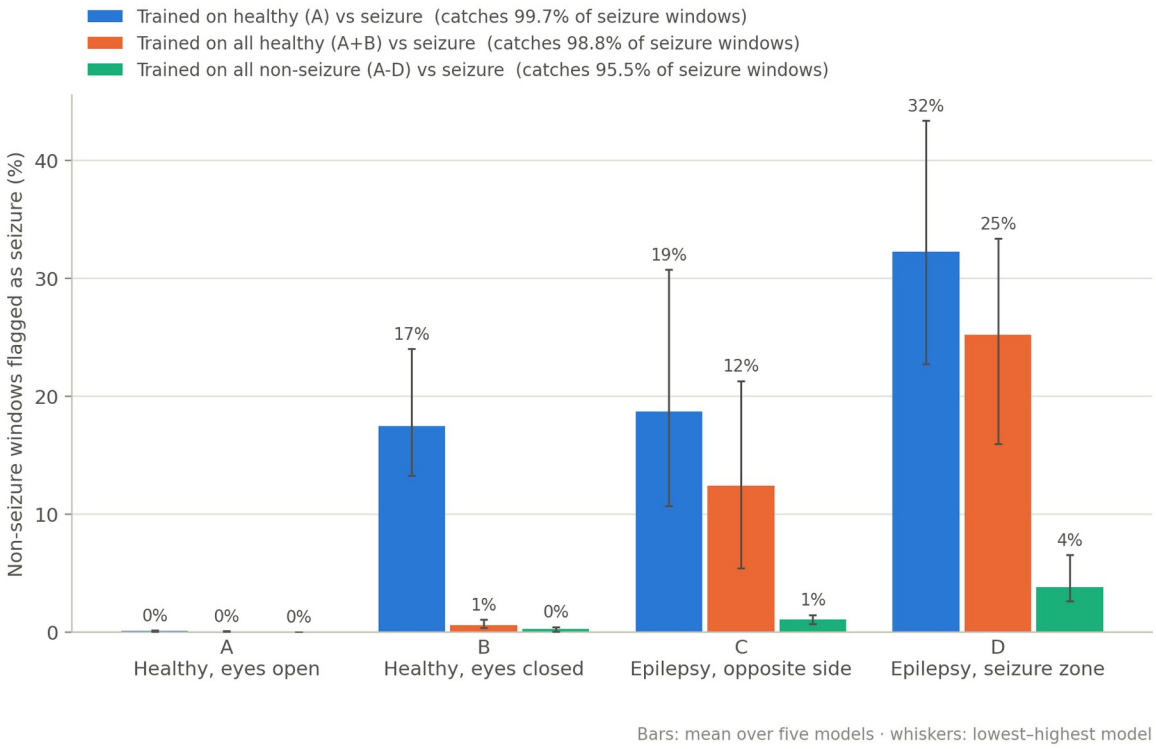


Figure 11. False-alarm rates on held-out non-seizure recordings for detectors trained with different non-seizure classes. Bars: mean of five classifiers; whiskers: range across classifiers.

Table 17. Detection and false-alarm rates (%) by training-set composition (mean of five classifiers).

Non-seizure training data	Detected (E)	A	B	C	D
A only	99.7	0.1	17.5	18.7	32.2
A + B	98.8	0.0	0.5	12.4	25.2
A + B + C + D	95.5	0.0	0.2	1.1	3.8

4.10 Effect of training-set size

On T2, balanced accuracy rose from 93.5% with 40 training recordings to 97.4% with 400 and plateaued beyond about 240 recordings. On T5 it rose from 56.8% with 16 recordings to 71.4% with 160 and had not plateaued (Figure 12), indicating that this task is limited by the amount of available data.

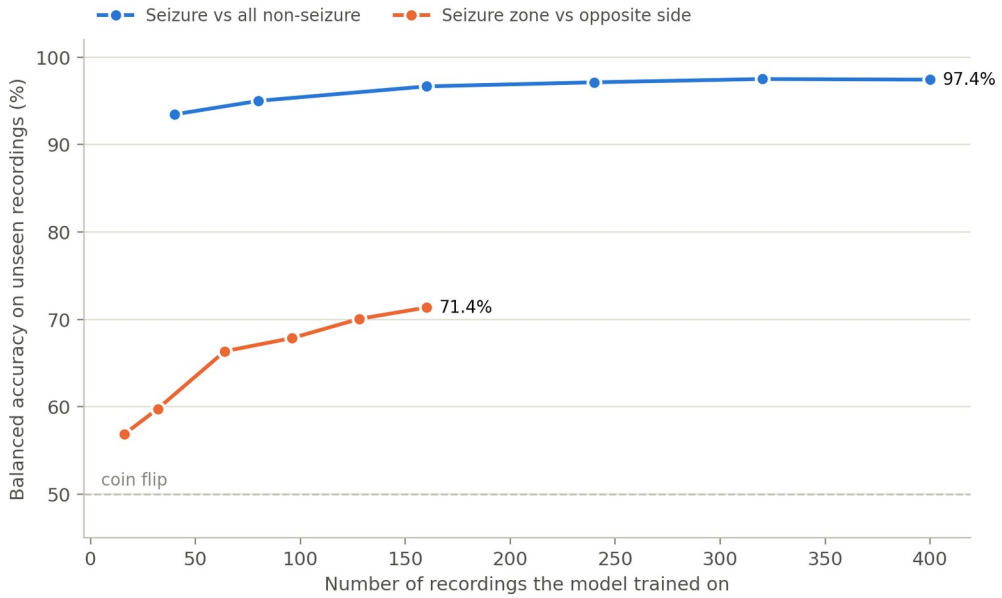


Figure 12. Learning curves: balanced accuracy on held-out recordings versus number of training recordings (gradient boosting; mean of 15 splits).

5. Discussion

5.1 Principal findings

Using a compact set of 18 interpretable features, all five classifiers recognized seizure activity in recordings unseen during training with 96.6–97.5% balanced accuracy, and aggregation over each recording raised this to 97.0–98.3%. Differences between classifiers were small on easier tasks and became meaningful only on the harder ones, where non-linear models had a consistent advantage.

5.2 What drives high accuracy

The ablation results show that, on the seizure-versus-healthy task, a single amplitude feature performs as well as the full models. Seizure EEG in this dataset is several times larger than all other EEG, so a high score on this task reflects that strong, simple difference rather than a detailed model of seizure dynamics. A practical implication is that simple baselines should be reported alongside complex models: they show how much of a result is explained by the most obvious feature.

5.3 Recording setup as a confounding factor

Healthy-versus-interictal classification (T4) reached 94.5–96.7%, and 94.8% with shape features alone. However, all healthy recordings were made with scalp electrodes and all patient recordings intracranially, and the two differ clearly in their spectra (Figure 4). The models may therefore have learned the recording setup rather than epilepsy-related activity, and this dataset cannot separate the two. Tasks that compare only intracranial recordings avoid this issue: seizure recognition within patients (T3) remained high, whereas discrimination of the epileptogenic zone (T5), the cleanest test of interictal epilepsy-related patterns here, was considerably harder.

5.4 Implications for evaluation

Two methodological choices had measurable effects. First, allowing windows from the same recording into both training and test sets inflated accuracy, most strongly on the hardest tasks, where the inflation is least likely to be noticed. Second, the composition of the non-seizure training data determined false-alarm behavior: a detector that has seen only one kind of normal EEG treats other normal EEG as abnormal. Grouped evaluation, diverse negative examples and reporting of sensitivity and false-alarm rates together give a more complete picture than accuracy alone.

5.5 Limitations

- **Small dataset.** 500 short segments from ten individuals in total.
- **Recording-level, not subject-level, evaluation.** Recordings are not linked to individuals, so the same person may appear in training and test data; generalization to new individuals is untested.
- **Recording-setup confound.** Healthy (scalp) and patient (intracranial) recordings differ in acquisition (Section 5.3).
- **Curated segments.** Single-channel, artifact-screened segments are much cleaner than continuous multi-channel clinical EEG.
- **No hyperparameter tuning.** Results compare methods under equal conditions but do not represent each model's best achievable performance.
- **Not clinically validated.** The models do not forecast seizures, do not distinguish a single seizure from epilepsy, and must not be used for diagnosis.

5.6 Future work

- Evaluate on a multi-subject, multi-channel dataset such as the CHB-MIT Scalp EEG Database, using leave-one-subject-out validation.
- Compare hand-crafted features with convolutional neural networks trained directly on raw EEG.
- Extend from segment classification to onset detection in continuous recordings.

6. Conclusion

This study built and evaluated a complete machine-learning pipeline for recognizing seizure activity in EEG. Standard classifiers trained on 18 features recognized seizure EEG with high accuracy on unseen recordings. The analyses also showed that this accuracy is driven largely by signal amplitude, that the recording setup can confound comparisons between healthy and patient EEG, and that evaluation protocol and training-data composition materially change measured performance. Subtler epilepsy-related patterns between seizures remain substantially harder to recognize and appear limited by the amount of data. Reporting such analyses alongside accuracy makes machine-learning results on EEG easier to interpret and to build upon.

Appendix A: Building a model on new data

The pipeline is not specific to the Bonn dataset. The repository includes a loader for any collection of single-channel recordings and a script that trains and evaluates all five classifiers with the protocol described in Section 3. The steps are:

1. **Organize the data.** Create one folder per class, with one recording per file. Each file holds one channel, as one value per line (.txt) or a column of a .csv file.
2. **Determine the sampling rate**, in samples per second, from the recording device or dataset documentation.
3. **Obtain the code** from the GitHub repository, or open the Google Colab notebook in a web browser.
4. **Run the training script** with the data folder and sampling rate. It filters, segments and extracts the 18 features, then trains and evaluates the five classifiers with recording-grouped cross-validation.
5. **Interpret the results** using balanced accuracy at window and recording level, compared with the chance level.
6. **Check for confounds.** Compare against a single-feature baseline, and verify that classes were not recorded under systematically different conditions.

```
my_data/
    seizure/    rec01.txt   rec02.txt   ...
    normal/     rec03.txt   rec04.csv   ...

python examples/use_your_own_data.py my_data 256
```

Recordings shorter than one window are skipped. At least ten recordings per class are recommended; as Figure 12 shows, harder problems continue to benefit from more data.

Appendix B: Glossary

Table 18. Glossary of terms.

Term	Definition
Balanced accuracy	Mean of per-class accuracies; robust to class imbalance
Band-pass filter	Filter that retains only frequencies within a chosen range
Contralateral	On the opposite side of the brain
Cross-validation	Repeated evaluation in which different subsets of data are held out for testing
Data leakage	Information from test data influencing training, which inflates performance
Epileptogenic zone	Brain region from which seizures originate
Feature	Numerical measurement describing an example
Ictal / interictal	During a seizure / between seizures
Inference	Applying a trained model to new data
Intracranial EEG	EEG recorded with electrodes inside the skull
Label	The known correct class of a training example
Overfitting	Fitting idiosyncrasies of training data at the expense of generalization
Permutation importance	Drop in performance when one feature's values are randomly shuffled
Sensitivity / specificity	True-positive rate / true-negative rate
Supervised learning	Learning from examples with known labels
Training	Fitting a model's parameters to labeled examples

Acknowledgments

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Use of AI tools

An AI assistant (Claude, by Anthropic) was used extensively in this project. Working under the author's direction, it wrote most of the analysis code, ran the initial experiments and drafted this manuscript. The author defined the project's goals and scope, ran the complete pipeline independently in Google Colab to reproduce the results, reviewed the findings and directed revisions of the text.

Data and code availability

The Bonn EEG dataset is available from Universitat Pompeu Fabra (upf.edu/web/ntsa/downloads). All code, result files and figures needed to reproduce this study are publicly available at https://github.com/sanjambahl/eeg-seizure-detection.

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